

Age and Stability Trade-offs in Remote Monitoring Systems

Nitya Sathyavageeswaran, Anand D. Sarwate, Roy D. Yates, Narayan Mandayam

Rutgers, The State University of New Jersey

Email: {nitya.s, anand.sarwate}@rutgers.edu, {ryates, narayan}@winlab.rutgers.edu

Abstract—Timely information is important in a wide variety of Internet of Things (IoT) services in which a shared server must manage two competing tasks: (i) processing a queue of jobs, and (ii) generating status updates to a remote monitor. This creates a fundamental trade-off between queue stability and data freshness. In this work, we model this scheduling decision as a Markov Decision Process (MDP) with the objective of minimizing a weighted sum of the average Age of Information (AoI) and the average queue length. We show that the optimal scheduling strategy is a queue-dependent age threshold which is monotonic. The shape of the switching curve differs according to different priority regimes. Finally, we compare the optimal MDP policy against heuristic policies.

I. INTRODUCTION

In Internet of Things (IoT) systems, the utility of collected data degrades rapidly over time, making timely information delivery critical [1]. As an example, consider an industrial application in which an installed device has a camera that is used to perform continuous high-speed quality control on an assembly line. Image processing is done locally and, to obtain high throughput, the local processor operates in a heavy-traffic regime. The camera must also transmit status updates to a monitor which is tracking the entire production line. These updates share the local processor, so generating them temporarily degrades the primary image-processing job pipeline to a slower service. The severity of this delay depends on the type of update being sent; transmitting a simple summary update may be fast, but processing a high resolution frame or doing extensive post processing may be slow.

In heavy traffic, even this brief interrupt for an update risks queue instability. This creates a fundamental trade-off between system stability and data freshness at the monitor. In this paper we use the Age of Information (AoI) [2] metric to quantify the freshness of information at the monitor. An *always-update* policy prioritizes AoI but leads to queue instability, whereas an *always-serve* policy stabilizes the queue but allows AoI to grow unbounded. Balancing these extremes requires a state-aware scheduling approach.

In this paper we formulate the problem of balancing AoI and stability as a Markov Decision Process (MDP) in which a local shared processor must dynamically choose whether to service a queue holding an incoming stream of data processing jobs or generate a fresh status update for the monitor. We use a weighted sum of the average age and the expected queue length as a cost function. We show that the optimal scheduling

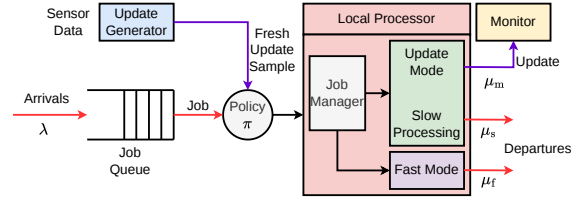


Fig. 1. System model for the status update and the queueing process.

policy follows a state-aware monotonic switching curve that adapts to different parameter regimes. We empirically compare the optimal policy to two simple heuristics: a Memoryless policy that randomly chooses to send an update, and a Myopic policy that minimizes the expected cost in the next time step. Because our optimal policy does not have a closed form expression, we identify a low-complexity heuristic that approximates the switching curve and demonstrates that it can achieve near-optimal performance.

Related work. Average age has been extensively studied for various queuing systems [2]–[5]. Prior work has examined scheduling strategies for minimizing long run average age in multi-user settings [6]–[8]. MDP-based formulations have also been used to derive optimal age-aware policies [9]–[13]. AoI minimization subject to throughput constraints has also been studied [6], [14]–[16]. In contrast, in our work the stability requirement arises not from an external throughput demand, but from a shared processor whose service rate degrades by generating updates, hence coupling stability and freshness. Recent works have also begun to incorporate service delays and system dynamics into the design of update policies. In particular, sampling and scheduling problems with stochastic service times and queuing effects have been studied [17]–[19].

II. SYSTEM MODEL AND PROBLEM FORMULATION

Notation. Random variables will be denoted by capital letters and realizations by lower case letters. Time passes in integer slots with slot $n \geq 0$ denoting the time interval $[n, n + 1)$.

We study a discrete time system consisting of a data source, a status update generator, a shared local processor, a queue, and a remote monitor as shown in Fig. 1. Within a slot n , the following events happen in order:

- 1) Arrivals: Data processing jobs arrive at a First-Come-First-Served (FCFS) queue. We model this as a Bernoulli

process with rate λ , where $X_n = 1$ if an arrival occurs in slot n , and $X_n = 0$ otherwise.

- 2) Scheduling: The scheduler observes the system state (queue length Q_n and age A_n), and chooses an action $U_n \in \{0, 1\}$.
- 3) Service and Delivery:
 - If $U_n = 0$ (*Serve Queue*), the local processor serves the queue *fast*. A job is successfully processed and departs with probability μ_f .
 - If $U_n = 1$ (*Generate Update*), the local processor pulls a fresh update from the source and spends one time slot attempting to deliver to the monitor. This new status update is successfully delivered at the end of the slot with probability μ_m . Concurrently, the processor serves the data queue at a *slow* rate; a job departs with probability $\mu_s \leq \mu_f$.

A. Problem Formulation

The scheduler determines the server's operating mode in each slot. We model the scheduler's actions over time as a sequence $\mathbf{U} = (U_1, U_2, \dots)$, where

$$U_n = \begin{cases} 1 & \text{Generate Update (Update Mode)} \\ 0 & \text{Serve Queue (Fast Mode).} \end{cases} \quad (1)$$

To use the AoI metric to measure timeliness, let $R_n = \max\{k \leq n : \text{update generated at } k \text{ delivered}\}$ be the generation time of the most recent update at the monitor. The age at the monitor at slot n is $A_n = n - R_n$. The age increases by 1 in every slot where a successful update does not occur. The average age is defined as¹

$$\Delta = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N (A_n + 1/2). \quad (2)$$

For the data processing job queue, an arrival and departure can occur in the same slot. The evolution of the queue backlog Q_n is governed by arrivals X_n and departures D_n :

$$Q_{n+1} = Q_n - D_n + X_n, \quad (3)$$

where the departure probability is given by

$$P(D_n = 1 | U_n, Q_n, X_n) = \begin{cases} \mu(U_n) & \text{if } Q_n + X_n > 0 \\ 0 & \text{if } Q_n + X_n = 0, \end{cases} \quad (4)$$

where $\mu(0) = \mu_f$ and $\mu(1) = \mu_s$. Similarly, we define the average queue length as:

$$\bar{Q} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N Q_n. \quad (5)$$

Our goal is to minimize the weighted sum of the average age and the average queue length. The average cost is defined as:

$$\bar{C} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N (A_n + \frac{1}{2} + wQ_n), \quad (6)$$

¹Actions occur slot by slot in discrete time but age is averaged continuously over the slot; when $A_n = a$, the average age over slot n is $a + 1/2$.

where $w > 0$ is a weight parameter representing the penalty for queue backlog relative to staleness.

Given these ingredients, we formulate the problem as a Markov Decision Process (MDP). We define the state of the system as $S_n = (Q_n, A_n)$. The MDP consists of a tuple $(\mathcal{S}, \mathcal{U}, P, C)$: $s = (q, a) \in \mathcal{S}$, where $q \in \mathbb{N}_0$ and $a \in \mathbb{N}$ are the countably infinite set of states, and $u \in \mathcal{U}$ is the action defined in (1).

We assume job queue arrivals are a rate λ Bernoulli process, independent of data service and update generation. The transition function $P((q', a') | (q, a), u)$ governs the probability of the transition $S_n \rightarrow S_{n+1}$. Since only one arrival or departure can happen per slot, the transition probabilities $P(q \rightarrow q' | u)$ conditioned on the action $U_n = u$ are given as follows

$$P(q \rightarrow q' | u) = \begin{cases} \lambda(1 - \mu(u)) & q' = q + 1 \\ (1 - \lambda)\mu(u)I_{\{q>0\}} & q' = q - 1 \\ 1 - \lambda(1 - \mu(u)) & q' = q \\ -(1 - \lambda)\mu(u)I_{\{q>0\}} & \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

Based on the chosen action, the joint state transitions are:

- If $u = 0$ (*Serve Queue*), the age increases by 1:

$$P((q', a + 1) | (q, a), 0) = P(q \rightarrow q' | 0). \quad (8)$$

- If $u = 1$ (*Generate Update*), the age resets to 1 with probability μ_m :

$$P((q', 1) | (q, a), 1) = \mu_m P(q \rightarrow q' | 1), \quad (9)$$

$$P((q', a + 1) | (q, a), 1) = (1 - \mu_m)P(q \rightarrow q' | 1). \quad (10)$$

We model the immediate cost C of taking action U in state S as

$$C_n = C(S_n, U_n) = A_n + \frac{1}{2} + wQ_n. \quad (11)$$

We want to find a policy f^* that minimizes the long term average cost given by

$$V_f(s) = \limsup_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}_f \left[\sum_{n=1}^N C(S_n, U_n) \middle| S_0 = s \right]. \quad (12)$$

III. HEURISTIC POLICIES

We first analyze a few heuristic policies to understand the age/queue length tradeoff.

A. Memoryless inspection

In this policy, the scheduler makes its decisions independently of the system state. In every slot, the scheduler chooses to generate a status update with fixed probability p . The average age at the monitor for this policy is given by

$$\Delta_{\text{mem}} = \frac{1}{p\mu_m} + \frac{1}{2}. \quad (13)$$

The detailed derivation of this average age is provided in the extended version [20]. The effective service rate, μ_{mem} for

this policy, which is the probability of a job departure from the queue in a slot is given by

$$\mu_{\text{mem}} = p\mu_s + (1-p)\mu_f. \quad (14)$$

The queue evolves as a discrete-time Geo/Geo/1 queue. Using standard results for discrete-time Markov queues [21], the steady-state average queue length is given by

$$\bar{Q}_{\text{mem}} = \frac{\lambda(1 - \mu_{\text{mem}})}{\mu_{\text{mem}} - \lambda}. \quad (15)$$

A fundamental requirement for the system is to maintain queue stability, ensuring that the average queue length $\bar{Q}$ remains finite. Given the arrival rate λ and the effective service rate μ_{mem} , the system is stable if and only if $\lambda < \mu_{\text{mem}}$. This implies that the update probability must satisfy

$$p < p_{\text{max}} \triangleq \frac{\mu_f - \lambda}{\mu_f - \mu_s}. \quad (16)$$

From (13), we see that the average age decreases as the update probability p increases. However, a higher p reduces the effective service rate which in turn increases the queue backlog. This competing dynamic implies there is a unique update probability p^* that optimally resolves this tradeoff, which is given in the following proposition.

Proposition 1. *For the Memoryless policy where updates are chosen with probability p in each slot, the probability p^* that minimizes the average cost $\bar{C}(p)$ is*

$$p^* = \min \left(1, \frac{\mu_f - \lambda}{\Delta\mu + \sqrt{w\lambda(1-\lambda)\mu_m\Delta\mu}} \right), \quad (17)$$

where $\Delta\mu = \mu_f - \mu_s$, $\lambda < \mu_f$. If $\lambda \geq \mu_f$, no stable policy exists.

The proof is provided in the extended version [20].

B. Myopic policy

In this policy the scheduler minimizes the expected cost in the next time slot. At any time n with state $S_n = (q, a)$, the scheduler selects the action $u \in \{0, 1\}$ that minimizes the one-step expected cost $E[C_{n+1} | S_n = (q, a), U_n = u]$.

By evaluating the expected next-step costs for updating versus serving the queue (detailed in the extended version [20]), we find that the Myopic scheduler behaves as a deterministic threshold policy. It generates a status update ($U_n = 1$) if the current age $A_n = a$ satisfies

$$a > \begin{cases} \frac{w\lambda(\mu_f - \mu_s)}{\mu_m} & \text{if } q = 0 \\ \frac{w(\mu_f - \mu_s)}{\mu_m} & \text{if } q > 0. \end{cases} \quad (18)$$

Otherwise, the scheduler chooses to serve the queue ($U_n = 0$). We denote the deterministic age threshold for a back-logged queue ($q > 0$) as $a_{\text{myopic}}^* = w(\mu_f - \mu_s)/\mu_m$.

To establish the stability of the Myopic policy, a standard approach would rely on Foster's stability criterion [22] using a Lyapunov function such as $L(Q_n, A_n) = Q_n$. Basic versions of this criterion require the expected one-step drift to be strictly

negative outside a finite set of states. However, under the Myopic Policy when $A_n > K$, the expected one-step queue drift, $\lambda - \mu_s$ is strictly greater than zero. Also the age process lacks autonomy since the threshold changes when $Q_n = 0$. To resolve this we adopt a methodology inspired by Yates et al. [23] and construct a fictitious autonomous age process. This lets us establish queue ergodicity using the Markov-modulated framework of Foss et al. [24].

Theorem 1 (Myopic Stability region). *Under the Myopic scheduling policy, the queue is stable (positive recurrent) if the arrival rate satisfies the following condition:*

$$\lambda < \frac{K\mu_f\mu_m + \mu_s}{K\mu_m + 1}, \quad (19)$$

where $K = \lfloor a_{\text{myopic}}^* \rfloor$ is the deterministic age threshold for a backlogged queue.

We construct a fictitious system applying a constant threshold K , letting age operate as an independent renewal cycle. By renewal reward theory, its effective service rate is

$$\mu_{\text{eff}} = \frac{K\mu_f + \mu_s/\mu_m}{K + 1/\mu_m} = \frac{K\mu_f\mu_m + \mu_s}{K\mu_m + 1}. \quad (20)$$

By Corollary 1 of Foss et al. [24], this independent system is stable if $\lambda < \mu_{\text{eff}}$. Since the true Myopic policy clears backlogged jobs at least as fast as the fictitious system, it is inherently stable (full proof in the extended version [20]).

IV. OPTIMAL POLICY ANALYSIS

We now turn to finding the optimal policy f^* which minimizes the long term average cost in (12). We first define the discounted cost $V_{f,\beta}(s)$ as follows:

$$V_{f,\beta}(s) = E_f \left[\sum_{t=0}^{\infty} \beta^t C(S_t, U_t) \middle| S_0 = s \right], \quad (21)$$

where $s = (q, a)$ is the initial system state and $\beta \in (0, 1)$ is the discount factor. We define the optimal discounted value function

$$V_{\beta}(s) = \inf_f V_{f,\beta}(s). \quad (22)$$

If the value function $V_{\beta}(s)$ is finite, it satisfies the following discounted cost optimality equation:

$$V_{\beta}(s) = \min_{u \in \{0,1\}} \left(C(s, u) + \beta \sum_{s'} P_{ss'}(u) V_{\beta}(s') \right), \quad (23)$$

for all $s \in \mathcal{S}$, where $P_{ss'}(u)$ is the transition probability from state s to s' under action u .

We can define a value iteration $V_{\beta,n}(s)$ recursively as

$$\begin{aligned} V_{\beta,n}(q, a) &= \min_{u \in \{0,1\}} \left(C((q, a), u) + \beta \sum_{s'} P_{ss'}(u) V_{\beta,n-1}(s') \right) \\ &= a + wq + \min \left(\beta E[V_{\beta,n-1}(Q'_0, a + 1)], \right. \\ &\quad \left. \beta\mu_m E[V_{\beta,n-1}(Q'_1, 1)] \right. \\ &\quad \left. + \beta(1 - \mu_m) E[V_{\beta,n-1}(Q'_1, a + 1)] \right), \end{aligned} \quad (24)$$

for any $n \geq 1$, with $V_{\beta,0}(s) = 0$ for all $s \in \mathcal{S}$ and $\beta \in (0, 1)$, where Q'_0 and Q'_1 denote the next-slot queue lengths under Fast Mode and Update Mode, respectively.

Under Sennott's conditions [25] (verified in the extended version [20]), there exists a differential cost function $h(s)$ and a stationary deterministic policy f^* satisfying the average cost optimality equation

$$g + h(s) = \min_{u \in \{0,1\}} \left(C(s, u) + \sum_{s'} P_{ss'}(u) h(s') \right), \quad (25)$$

for all $s \in \mathcal{S}$, where g is the minimum average cost.

Remark 1 (Stability of the Optimal Policy). Since the immediate cost function explicitly penalizes the queue length ($C(S_n, U_n) = A_n + 1/2 + wQ_n$ with $w > 0$), any policy that yields a finite average cost must inherently maintain a finite average queue length. In Section III-A we see that whenever $\lambda < \mu_f$, a Memoryless policy with update probability $p < p_{\max}$ stabilizes the queue and achieves a finite average cost. By definition, the optimal policy f^* achieves an average cost no greater than this Memoryless baseline. Therefore, the optimal policy guarantees a finite average queue length, and ensures strict system stability for any $\lambda < \mu_f$.

A. Relative Value Iteration (RVI)

To numerically evaluate the optimal policy f^* , we truncate the state space to a finite grid $\mathcal{S}_{\text{trunc}} = \{0, \dots, Q_{\max}\} \times \{1, \dots, A_{\max}\}$. The optimal policy for the truncated MDP converges to that of the original MDP as $Q_{\max}, A_{\max} \rightarrow \infty$, provided that the truncated MDP is *unichain*, and the conditions of Sennott [25] hold (Verified in the extended version [20]). Our truncated MDP satisfies this unichain condition because, under any stationary policy, the Markov chain consists of a single recurrent class. The state $(0, 1)$ can be reached from all other states $(q, a) \in \mathcal{S}_{\text{trunc}}$. This is because, from any state, there is a strictly positive probability of having a finite sequence of zero arrivals coupled with consecutive fast service completions, followed by a successful update.

We get the exact optimal policy over $\mathcal{S}_{\text{trunc}}$ by applying RVI [26]. The value function is initially set to $V_0(s) = 0$ for all states $s \in \mathcal{S}_{\text{trunc}}$. At each iteration $n \geq 0$, we update:

$$V_{n+1}(s) = \min_u [C(s, u) + E[V_n(s')]] - V_n(s_{\text{ref}}), \quad (26)$$

where $s_{\text{ref}} = (0, 1)$ is a fixed reference state. The iteration stops when the condition $\max_{s \in \mathcal{S}_{\text{trunc}}} |V_{n+1}(s) - V_n(s)| < \epsilon$ is satisfied.

Fig. 2 and Fig. 3 illustrate the exact optimal scheduling policy derived via RVI over the state space for three different values of the queue penalty weight $w \in \{0.1, 1, 10\}$. The optimal policy is a non-decreasing switching curve $a^*(q)$ which partitions the state space into two distinct operating regions. The blue region below the boundary indicates states where the optimal action is to serve the queue in Fast Mode ($u = 0$) to clear backlog, and the red region above the boundary indicates states where the optimal action is to generate an update ($u = 1$) to prioritize data freshness. As the penalty for queue backlog

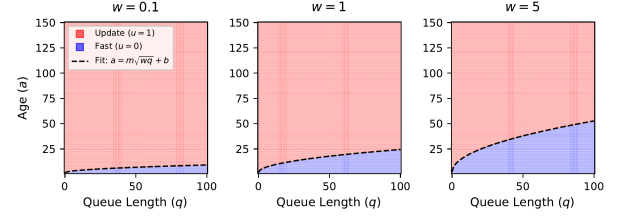


Fig. 2. The optimal switching boundaries $a^*(q)$ for varying queue penalty weights $w \in \{0.1, 1, 5\}$. For any given weight curve, the region above the boundary indicates the Update Mode ($u = 1$), while the region below indicates the Fast Mode ($u = 0$). The system parameters are fixed at $\lambda = 0.4$, $\mu_f = 0.5$, $\mu_s = 0.3$, and $\mu_m = 0.8$.

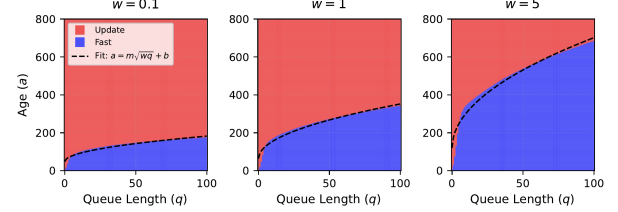


Fig. 3. The optimal switching boundaries $a^*(q)$ for varying queue penalty weights $w \in \{0.1, 1, 5\}$. For any given weight curve, the region above the boundary indicates the Update Mode ($u = 1$), while the region below indicates the Fast Mode ($u = 0$). The system parameters are fixed at $\lambda = 0.4$, $\mu_f = 0.5$, $\mu_s = 0.3$, and $\mu_m = 0.008$.

w increases, this monotonic boundary (age threshold) shifts upward, which indicates the scheduler's hesitation to interrupt queue service for status updates.

Note: The dashed lines in Fig. 2 and Fig. 3 represent a low-complexity structural heuristic that approximates this boundary, which we introduce and analyze in Section IV-B.

B. Low-Complexity Square-Root Heuristic

While the exact MDP formulation yields an optimal policy f^* , storing an infinite-sized lookup table is impractical for implementation on resource-constrained edge devices. Our empirical evaluations using RVI reveal that the optimal switching boundary exhibits a concave, square root shape in the (q, a) state space. When the scheduler prioritizes to send updates to the monitor, the queue penalty grows proportionally to wq . Postponing updates, and prioritizing serving the queue jobs makes the age build up, and the cumulative age penalty, over that time period grows quadratically in a . Balancing a quadratic age-growth term a^2 against a linear queue term wq yields the square-root switching boundary.

We define the Square-Root Heuristic, denoted as f_{sqr}^* , which maps the current system state $S_n = (q, a)$ to an action $u \in \{0, 1\}$ as follows:

$$f_{\text{sqr}}^*(q, a) = \begin{cases} 1 & \text{if } a > m\sqrt{wq} + b \\ 0 & \text{otherwise,} \end{cases} \quad (27)$$

where $m > 0$ and $b \geq 0$ are tunable parameters.

Here, b is the baseline age tolerance of the system at $q = 0$, and m governs sensitivity to queue backlog. A larger m

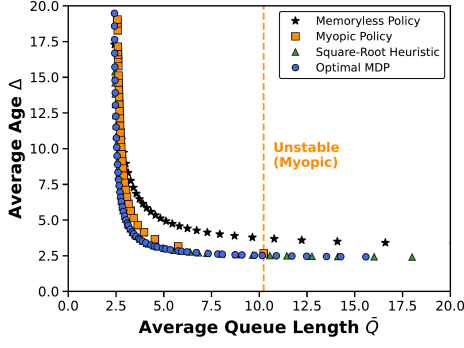


Fig. 4. Performance comparison of the Optimal MDP, Square-Root heuristic, Myopic, and Memoryless policies. The Pareto frontiers are generated by varying the queue penalty weight w for the Optimal MDP, the deterministic age threshold K for the Myopic policy, and the update probability p for the Memoryless policy. System parameters $\lambda = 0.4$, $\mu_f = 0.5$, $\mu_s = 0.3$, and $\mu_m = 0.8$.

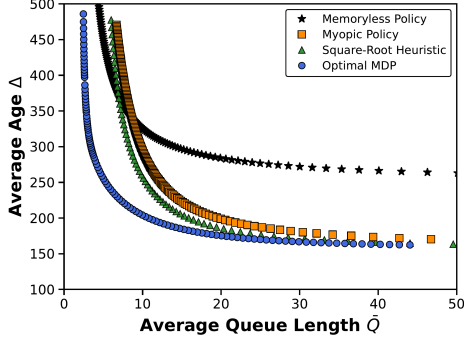


Fig. 5. Performance comparison of the Optimal MDP, Square-Root heuristic, Myopic, and Memoryless policies under a severely constrained update channel. System parameters $\lambda = 0.4$, $\mu_f = 0.5$, $\mu_s = 0.3$, and $\mu_m = 0.008$.

dictates a steeper initial penalty for the queued jobs, delaying status updates to prioritize queue stability. As shown by the dashed lines in Fig. 2, and Fig. 3, this heuristic uses just two parameters to approximate the optimal policy, avoiding the need for the infinite-size lookup table. The optimal parameters (m^*, b^*) can be determined offline by applying ordinary least squares regression to the exact MDP switching boundary. Alternatively, for systems where computing the full MDP is computationally prohibitive, they can be found directly via a low-dimensional grid search over the target stability region.

We compare optimal MDP policy and our square-root heuristic against two simpler heuristic baselines: the Memoryless policy and the Myopic policy. Fig. 4 and Fig. 5 illustrate the trade-off between the average queue length $\bar{Q}$ and the average Age of Information Δ at the monitor under fast ($\mu_m = 0.8$) and severely constrained ($\mu_m = 0.008$) update channels. The optimal MDP curve is generated by using RVI across varying queue penalty weights w , establishing the best achievable Pareto frontier. As w increases, the MDP heavily penalizes buffered jobs, minimizing $\bar{Q}$ while allowing Δ to grow. Conversely, as w approaches zero, the policy prioritizes data freshness, minimizing Δ at the cost of a larger queue.

For the Memoryless policy, the average age and queue length are plotted using the exact analytical expressions derived in (13) and (15). The update probability p is varied continuously from 0 up to the stability limit $p_{\max}$. As p increases, the system generates updates more frequently, and the average age reduces. However, because the policy makes scheduling decisions independently of the current queue backlog or data staleness, it yields the worst performance.

For the Myopic policy, the average age and queue length are evaluated numerically by computing the stationary distribution of the discrete-time Markov chain induced by the deterministic age threshold K . As $w \rightarrow 0$, the threshold K reduces, forcing the system to spend more time in the slow Update Mode ($\mu_s = 0.3$). Since $\lambda = 0.4 > \mu_s$, the effective service capacity falls below the stability bound from Theorem 1. The vertical dashed line in Fig. 4 marks this theoretical breaking point, beyond which the Myopic queue grows unbounded.

The impact of the update rate μ_m is stark when comparing the two regimes. Under fast updates (Fig. 4), the Square-Root heuristic achieves near-optimal performance, overlapping the MDP curve almost entirely. However, in the severely constrained environment (Fig. 5), an update takes an expected $1/\mu_m = 125$ slots, and a performance gap emerges between the Square-Root heuristic and the Optimal MDP. As seen in Fig. 3 for $w = 5$, the optimal policy remains highly update-averse to prevent queue overflow adopting a boundary that is offset from the heuristic. Also, the heuristic becomes infeasible for queue lengths below a certain threshold. This highlights the need of an exact, state-aware scheduling policy when managing extreme communication bottlenecks.

V. CONCLUSION

We studied the fundamental trade-off between data freshness and queue stability in remote monitoring IoT systems. We formulated the scheduling problem as a Markov Decision Process, and showed that the optimal policy for minimizing a weighted sum of Age of Information (AoI) and queue length follows a state-aware, monotonic switching curve. Our numerical evaluations revealed that this optimal boundary is well-approximated by a low-complexity square-root heuristic, unless the update channel is severely constrained. Furthermore, we compared the optimal approach against Memoryless and Myopic baselines, illustrating that state-aware scheduling not only improves the Pareto frontier but also drastically expands the stable operating region of the system.

ACKNOWLEDGEMENTS

This material is based upon work supported by the National Science Foundation under grant CNS-2148104 and is supported in part by funds from federal agency and industry partners as specified in the Resilient & Intelligent NextG Systems (RINGS) program.

REFERENCES

- [1] S. Shukla, M. F. Hassan, D. C. Tran, R. Akbar, I. V. Paputungan, and M. K. Khan, "Improving latency in internet-of-things and cloud computing for real-time data transmission: a systematic literature review (slr)," *Cluster Computing*, vol. 26, no. 5, pp. 2657–2680, 2023.
- [2] S. Kaul, R. Yates, and M. Gruteser, "Real-time status: How often should one update?" in *2012 Proceedings IEEE INFOCOM*. IEEE, 2012, pp. 2731–2735.
- [3] C. Kam, S. Kompella, G. D. Nguyen, J. E. Wieselthier, and A. Ephremides, "On the age of information with packet deadlines," *IEEE Transactions on Information Theory*, vol. 64, no. 9, pp. 6419–6428, 2018.
- [4] C. Kam, S. Kompella, G. D. Nguyen, and A. Ephremides, "Effect of message transmission path diversity on status age," *IEEE Transactions on Information Theory*, vol. 62, no. 3, pp. 1360–1374, 2016.
- [5] R. D. Yates and S. K. Kaul, "The age of information: Real-time status updating by multiple sources," *IEEE Transactions on Information Theory*, vol. 65, no. 3, pp. 1807–1827, 2018.
- [6] I. Kadota, A. Sinha, and E. Modiano, "Optimizing age of information in wireless networks with throughput constraints," in *IEEE INFOCOM 2018-IEEE Conference on Computer Communications*. IEEE, 2018, pp. 1844–1852.
- [7] Y. Sun, E. Uysal-Biyikoglu, and S. Kompella, "Age-optimal updates of multiple information flows," in *IEEE INFOCOM 2018-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. IEEE, 2018, pp. 136–141.
- [8] Y.-P. Hsu, E. Modiano, and L. Duan, "Scheduling algorithms for minimizing age of information in wireless broadcast networks with random arrivals," *IEEE Transactions on Mobile Computing*, vol. 19, no. 12, pp. 2903–2915, 2020.
- [9] Y.-P. Hsu, "Age of information: Whittle index for scheduling stochastic arrivals," in *2018 IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2018, pp. 2634–2638.
- [10] E. T. Ceran, D. Gündüz, and A. György, "Average age of information with hybrid arq under a resource constraint," *IEEE Transactions on Wireless Communications*, vol. 18, no. 3, pp. 1900–1913, 2019.
- [11] H. Tang, J. Wang, L. Song, and J. Song, "Minimizing age of information with power constraints: Multi-user opportunistic scheduling in multi-state time-varying channels," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 5, pp. 854–868, 2020.
- [12] A. M. Bedewy, Y. Sun, S. Kompella, and N. B. Shroff, "Optimal sampling and scheduling for timely status updates in multi-source networks," *IEEE Transactions on Information Theory*, vol. 67, no. 6, pp. 4019–4034, 2021.
- [13] N. Sathyavageeswaran, R. D. Yates, A. D. Sarwate, and N. M. Rutgers, "Timely offloading in mobile edge cloud systems," in *2024 IEEE Information Theory Workshop (ITW)*. IEEE, 2024, pp. 133–138.
- [14] I. Kadota, A. Sinha, and E. Modiano, "Scheduling algorithms for optimizing age of information in wireless networks with throughput constraints," *IEEE/ACM Transactions on Networking*, vol. 27, no. 4, pp. 1359–1372, 2019.
- [15] J. Sun, L. Wang, Z. Jiang, S. Zhou, and Z. Niu, "Age-optimal scheduling for heterogeneous traffic with timely throughput constraints," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 5, pp. 1485–1498, 2021.
- [16] N. Lu, B. Ji, and B. Li, "Age-based scheduling: Improving data freshness for wireless real-time traffic," in *Proceedings of the eighteenth ACM international symposium on mobile ad hoc networking and computing*, 2018, pp. 191–200.
- [17] T. Z. Ornee and Y. Sun, "Sampling for remote estimation through queues: Age of information and beyond," in *2019 International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOPT)*. IEEE, 2019, pp. 1–8.
- [18] Y. Sun, Y. Polyanskiy, and E. Uysal-Biyikoglu, "Remote estimation of the wiener process over a channel with random delay," in *2017 IEEE International Symposium on Information Theory (ISIT)*, 2017, pp. 321–325.
- [19] Y. Sun, E. Uysal-Biyikoglu, R. D. Yates, C. E. Koksai, and N. B. Shroff, "Update or wait: How to keep your data fresh," *IEEE Transactions on Information Theory*, vol. 63, no. 11, pp. 7492–7508, 2017.
- [20] N. Sathyavageeswaran, A. D. Sarwate, R. D. Yates, and N. Mandayam, "Age and stability trade-offs in remote monitoring systems," *arXiv preprint arXiv:2606.15746*, 2026.
- [21] S. M. Ross, *Introduction to probability models*. Academic press, 2014.
- [22] F. G. Foster, "On the stochastic matrices associated with certain queuing processes," *The Annals of Mathematical Statistics*, vol. 24, no. 3, pp. 355–360, 1953.
- [23] R. D. Yates and H. Mahdavi-Doost, "Energy harvesting receivers: Packet sampling and decoding policies," *IEEE Journal on Selected Areas in Communications*, vol. 33, no. 3, pp. 558–570, 2015.
- [24] S. Foss, S. Shneer, and A. Tyurlikov, "Stability of a markov-modulated markov chain, with application to a wireless network governed by two protocols," *Stochastic Systems*, vol. 2, no. 1, pp. 208–231, 2013.
- [25] L. I. Sennott, "Average cost optimal stationary policies in infinite state markov decision processes with unbounded costs," *Operations Research*, vol. 37, no. 4, pp. 626–633, 1989.
- [26] M. L. Puterman, "Markov decision processes: Discrete stochastic dynamic programming," 1994.